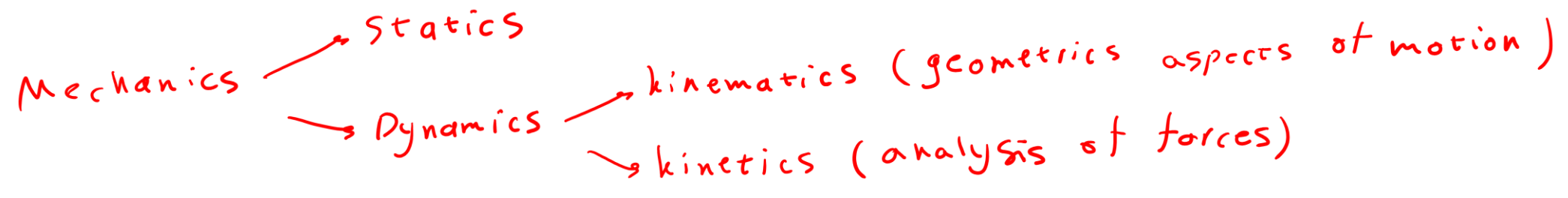


Kinematics



$$v_{avg} = \frac{\Delta s}{\Delta t}$$

$$\Delta s \longrightarrow ds$$

$$v = \frac{ds}{dt}$$

$$a_{avg} = \frac{\Delta v}{\Delta t}$$

$$\Delta t \longrightarrow dt$$

$$\Delta v \longrightarrow dv$$

$$a = \frac{dv}{dt}$$

$$s = s_0 + v_0 t + \frac{1}{2} a t^2$$

$$v = v_0 + a t$$

$$v^2 = v_0^2 + 2a(s - s_0) \quad \xrightarrow{\quad} \quad a ds = v dv$$

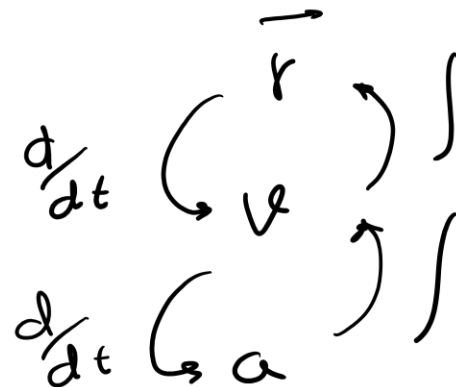
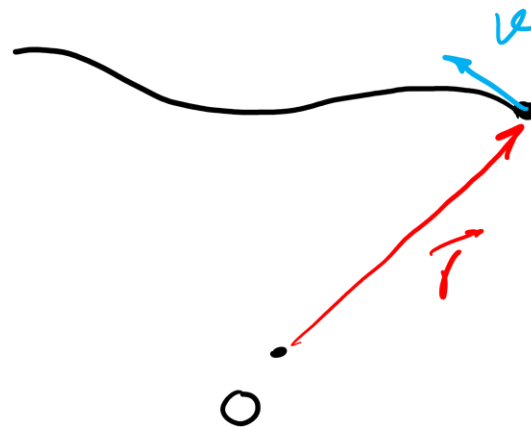
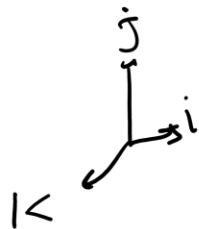
$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d(x\vec{i})}{dt} + \frac{d(y\vec{j})}{dt} + \frac{d(z\vec{k})}{dt}$$

$$= \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} + \frac{dz}{dt}\vec{k}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

$$\frac{d(x\vec{i})}{dt} = \frac{dx}{dt}\vec{i} + x\frac{d\vec{i}}{dt}$$



$$\vec{r} = r \cos(\omega t) \vec{i} + r \sin(\omega t) \vec{j}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = -r\omega \sin(\omega t) \vec{i} + r\omega \cos(\omega t) \vec{j}$$

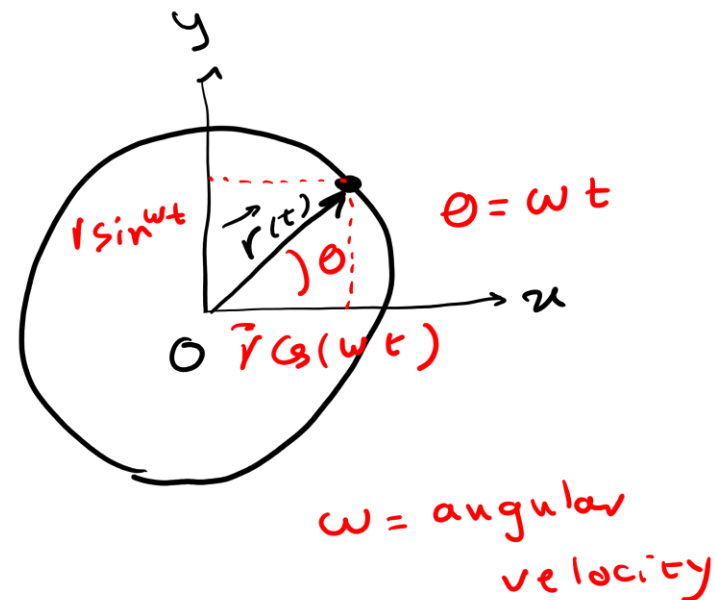
$$v = |\vec{v}| = \sqrt{r^2 \omega^2 \sin^2(\omega t) + r^2 \omega^2 \cos^2(\omega t)} = \sqrt{r^2 \omega^2 (\sin^2(\omega t) + \cos^2(\omega t))}$$

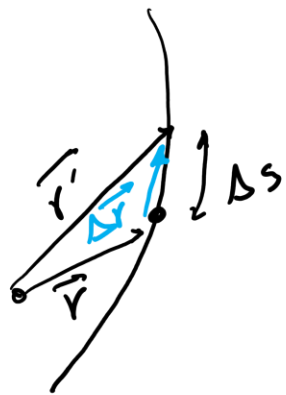
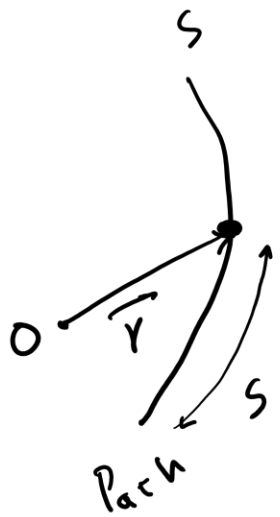
$$v = \text{speed} = \sqrt{r^2 \omega^2} = r\omega$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = -r\omega^2 \cos(\omega t) \vec{i} - r\omega^2 \sin(\omega t) \vec{j}$$

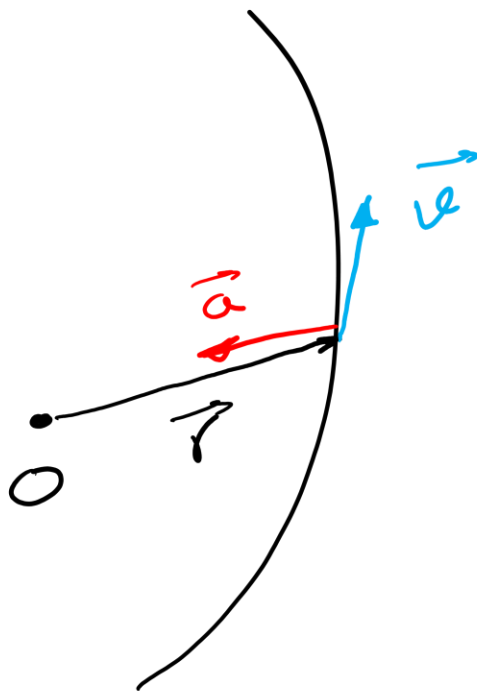
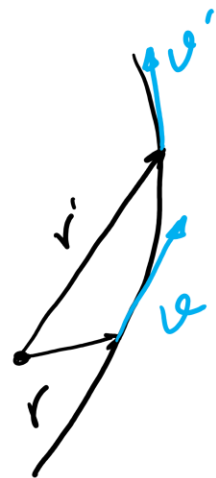
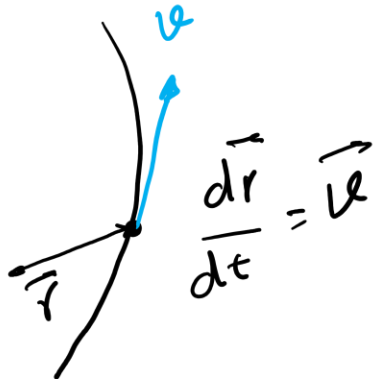
$$\vec{a} = -\omega^2 (r \cos(\omega t) \vec{i} + r \sin(\omega t) \vec{j})$$

$$\vec{a} = -\omega^2 \vec{r}$$

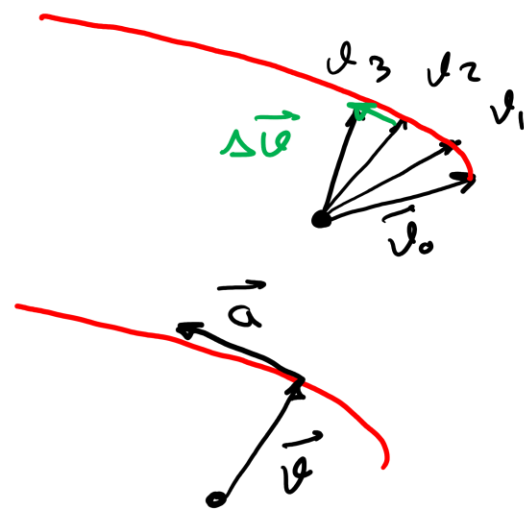




$\Delta t \longrightarrow dt$



Hodograph

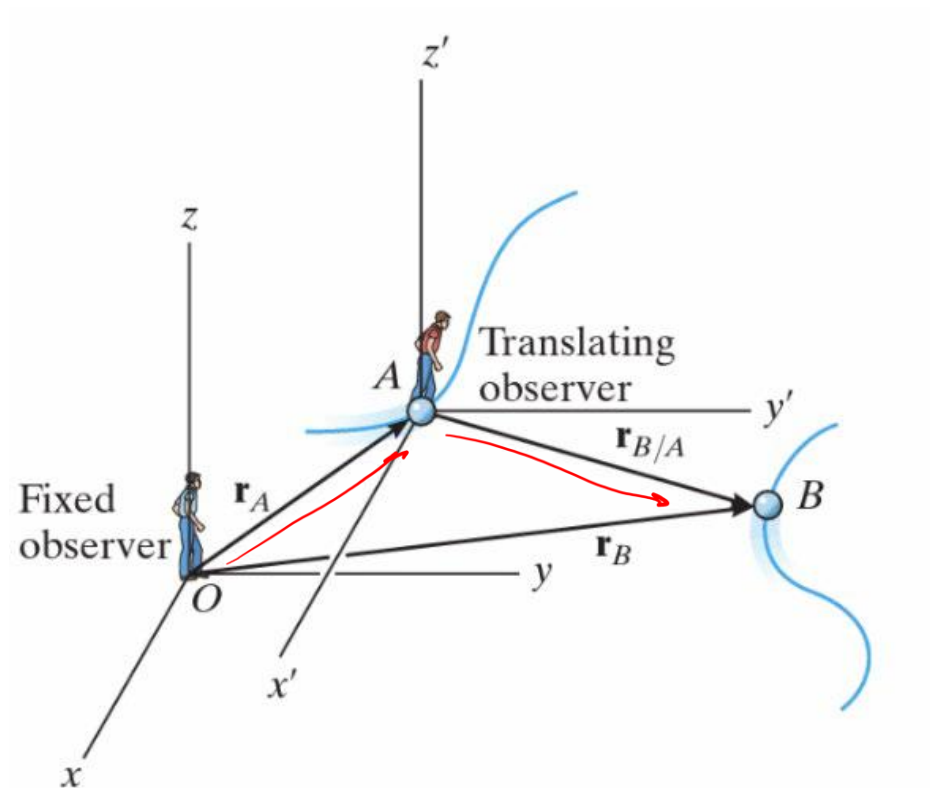


Relative motion

$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$$

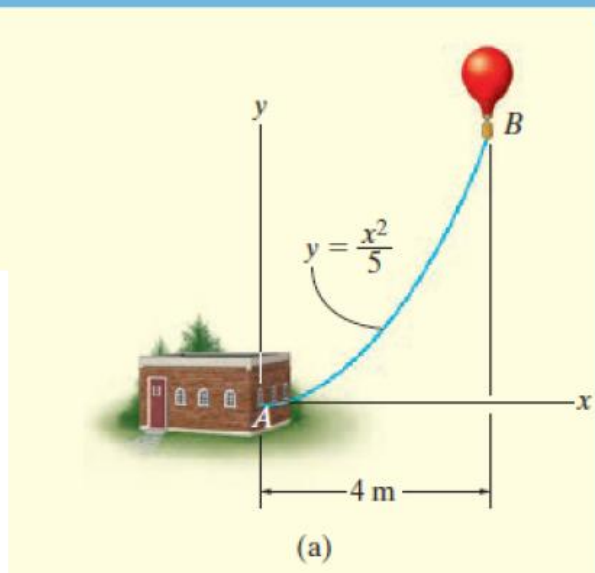
$$\vec{u}_B = \vec{u}_A + \vec{u}_{B/A}$$

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$



EXAMPLE 12.9

At any instant the horizontal position of the weather balloon in Fig. 12-18a is defined by $x = (2t)$ m, where t is in seconds. If the equation of the path is $y = x^2/5$, determine the magnitude and direction of the velocity and the acceleration when $t = 2$ s.



$$x = 2t$$

$$v_x = \dot{x} = \frac{d(2t)}{dt} = 2 \text{ m/s}$$

$$v_y = \dot{y} = \frac{d(x^2/5)}{dt} = \frac{1}{5} 2x \dot{x} = \frac{1}{5} 2 \times 4 \times 2 = 3.2 \text{ m/s}$$

$$v = \sqrt{2^2 + 3.2^2} = 3.77 \text{ m/s}$$

$$\theta = \tan^{-1} \frac{v_y}{v_x} = \tan^{-1} \frac{3.2}{2} = 58.0^\circ$$

$$a_x = 0, \quad a_y = \ddot{y} = \frac{d}{dt} (2x\dot{x}/5) = \frac{2}{5} (\dot{x}\dot{x} + x\ddot{x})$$

$$\frac{2}{5} (2 \times 2) + 4 \times 0$$

$$x' \dot{x} \ddot{x} \frac{dx}{dt}$$

$$y = u^n$$

$$y' = n \dot{u} u^{n-1}$$

$$u = 2t$$

$$\dot{u} = 2$$

$$\ddot{u} = 0$$

$$u = 2t$$

$$t = 2 \rightarrow u = 4$$

$$y = \frac{u^2}{5} = \frac{4t^2}{5}$$

$$y' = \frac{4}{5} 2t, \quad t = 2$$

$$y'' = \frac{8}{5}$$

$$y' = \frac{16}{5} = 3.2$$

Example 2

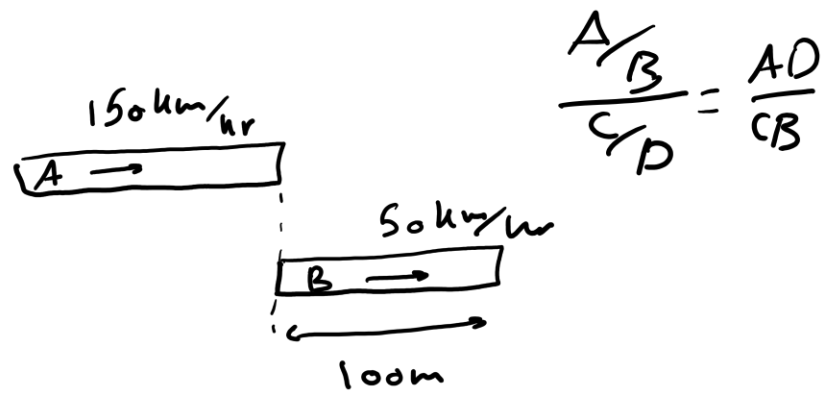
a) How long does it take the driver of a train (train-A), moving at a constant speed of 150 km.h⁻¹ to pass a train (train-B) of length 100 m that is travelling at a constant speed of 50 km.h⁻¹?

b) If a ball is dropped from the train-A, and frictional forces cause the ball to decelerate, in which direction does the ball move relative to:

- i. the train?
- ii. a signal point along the track?

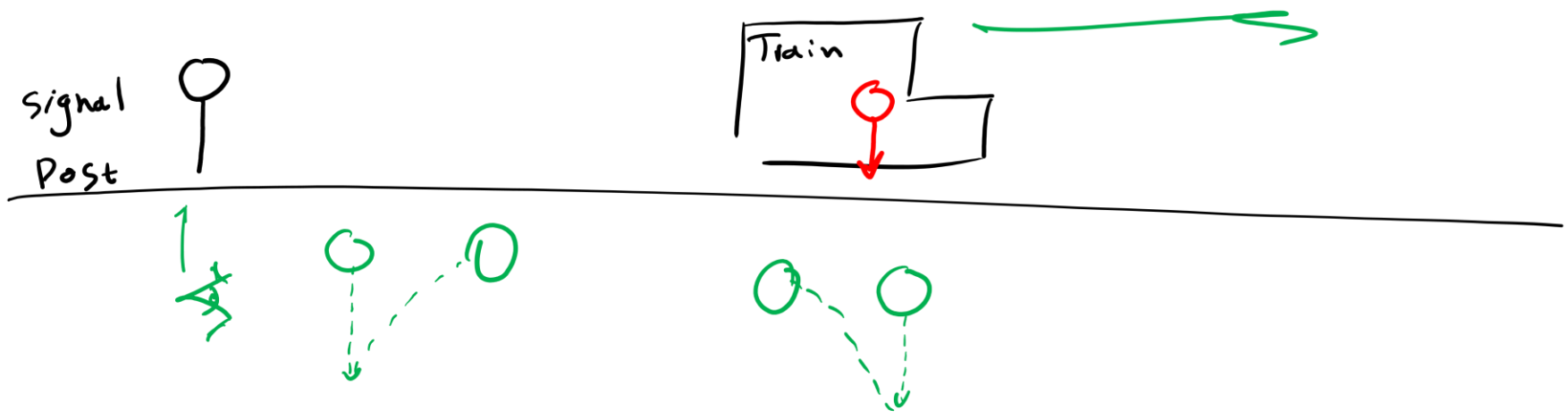
Does the acceleration differ between i. and ii.?

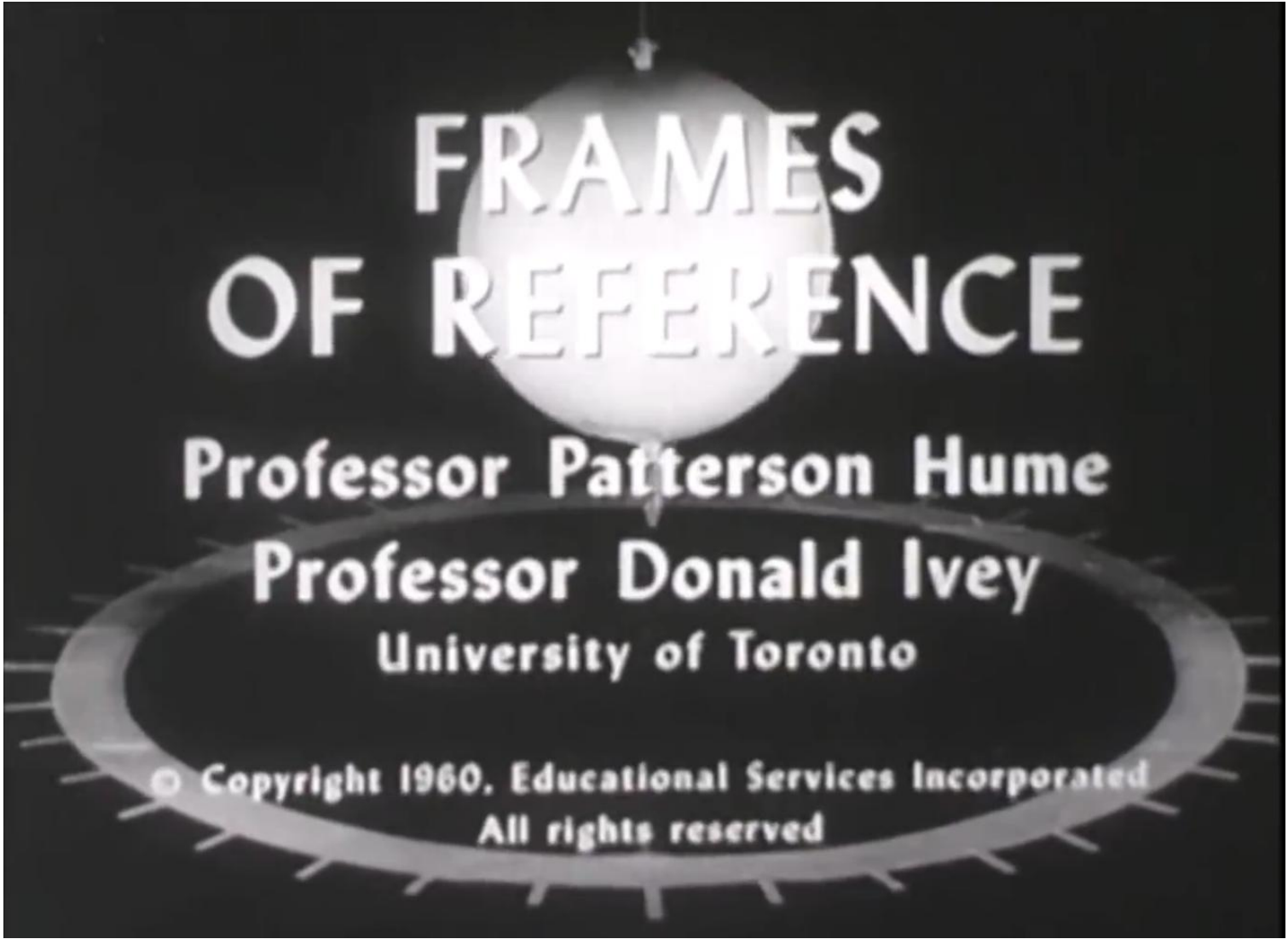
c) What does Inertia mean?



$$v = 150 - 50 = 100 \text{ km/hr}$$
$$= \frac{100 \times 10^3}{3600} \text{ m/s}$$

$$\Delta t = \frac{\Delta r}{v} = \frac{100 / 1}{\frac{100 \times 10^3}{3600}}$$
$$= 3.6 \text{ s}$$





FRAMES OF REFERENCE

Professor Patterson Hume

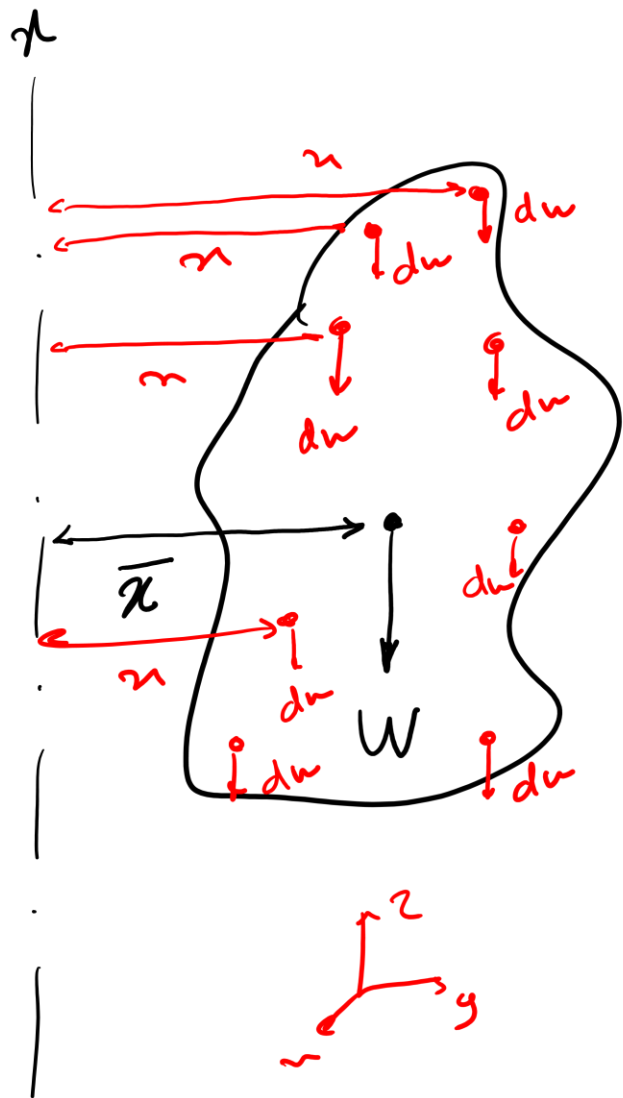
Professor Donald Ivey

University of Toronto

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Source: <https://www.youtube.com/watch?v=bJMYoj4hHqU>

Centre of Gravity



$$\int dw = w$$

$$\bar{x} w = \int x dw$$

$$\bar{x} = \frac{\int x dw}{w}$$

$$\bar{y} = \frac{\int y dw}{w}$$

$$\bar{z} = \frac{\int z dw}{w}$$

centre of gravity

$$W = mg \quad m = \rho V$$

$$g = \text{cte.}$$

$$\bar{x} = \frac{\int x g dm}{mg} \Rightarrow \bar{x} = \frac{\int x dm}{m}$$

$$\bar{y} = \frac{\int y dm}{m}$$

$$\bar{z} = \frac{\int z dm}{m}$$

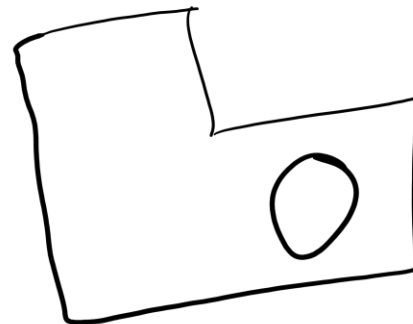
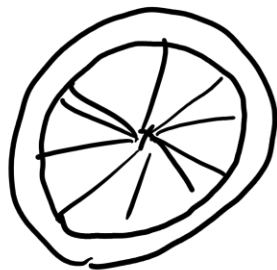
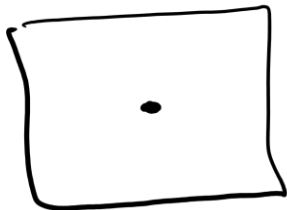
Centre of mass

$$\bar{x} = \frac{\int x \rho dV}{\rho V} = \frac{\int x dV}{V}$$

$$\bar{y} = \frac{\int y dV}{V}$$

$$\bar{z} = \frac{\int z dV}{V}$$

Centroid



$$\hat{\mu} = \frac{\sum \bar{x} w}{\sum w}$$

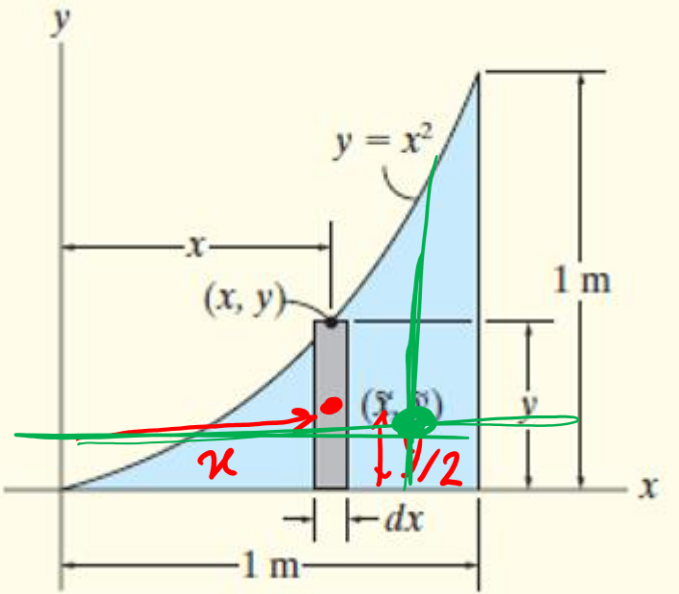
$$\hat{y} = \frac{\sum \bar{y} w}{\sum w}$$

$$\hat{z} = \frac{\sum \bar{z} w}{\sum w}$$

$$\hat{\mu} = \frac{\sum \bar{x} v}{\sum v}$$

EXAMPLE 6.1

Locate the centroid of the area shown in Fig. 6-5a.



$$\bar{x} = x$$

$$\bar{y} = y/2$$

$$A = \int dA$$

$$dA = y \, dx$$

$$\bar{x} = \frac{\int \bar{x} \, dA}{A}$$

$$\bar{y} = \frac{\int \bar{y} \, dA}{A}$$

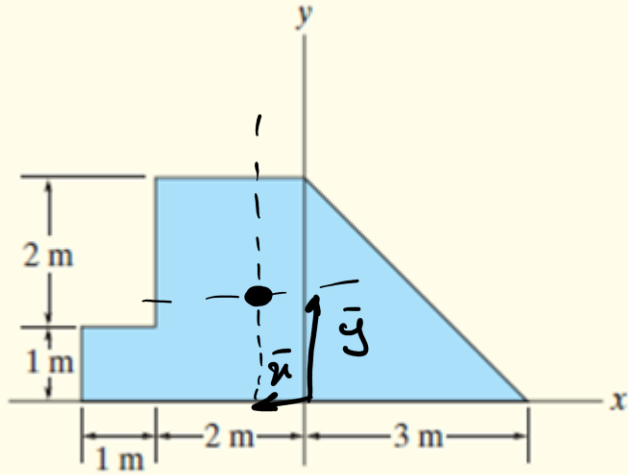
$$\bar{x} = \frac{\int \bar{x} \, dA}{A} = \frac{\int_0^1 x y \, dx}{\int_0^1 y \, dx} = \frac{\int_0^1 x^3 \, dx}{\int_0^1 x^2 \, dx} = \frac{0.250}{0.333} = 0.75 \, \text{m}$$

$$\bar{y} = \frac{\int \bar{y} \, dA}{\int dA} = \frac{\int_0^1 \frac{y}{2} y \, dx}{0.333} = \frac{\int_0^1 \frac{x^2 x^2}{2} \, dx}{0.333} = \frac{\frac{1}{2} \int_0^1 x^4 \, dx}{0.333} = \frac{0.1}{0.333}$$

$$\bar{y} = 0.3 \, \text{m}$$

EXAMPLE 6.4

Locate the centroid of the plate area shown in Fig. 6-9a.



Segment	Area	$\tilde{x}$	$\tilde{y}$	$\tilde{x}A$	$\tilde{y}A$
1	$\frac{1}{2} 3 \times 3 = 4.5$	1	1	4.5	4.5
2	$3 \times 3 = 9$	-1.5	1.5	-13.5	13.5
3	$-2 \times 1 = -2$	-2.5	2	5	-4
$\Sigma A = 11.5$				$\Sigma \tilde{x}A = -4$	$\Sigma \tilde{y}A = 14$

$$\bar{x} = \frac{\Sigma \tilde{x}A}{\Sigma A} = \frac{-4}{11.5} = -0.348 \text{ m}$$

$$\bar{y} = \frac{\Sigma \tilde{y}A}{\Sigma A} = \frac{14}{11.5} = 1.22 \text{ m}$$

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